

# UrjaMitra AI Architecture

The Ultimate AI Energy Twin & Machine Health Platform!



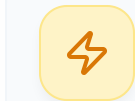
**Zero**  
DISRUPTION



**ZED**  
COMPLIANT



**STAGE 1 : SENSE**  
**Factory Hardware**



**Clip-on Power Sensors!**  
Zero-wire-cutting, snap-on current sensors instantly track every single watt your machines use.



**Machine Vitals**  
Vibration and temperature sensors attach magnetically to feel the hidden "health" of the motor!

- SPLIT-CORE CT
- NON-INVASIVE
- LEGACY READY



**STAGE 2 : CONNECT**  
**IoT Smart Hub**



**Instant Data Cleaning!**  
The edge chip instantly removes electrical noise to calculate true power consumption on the spot.



**Ultra-Secure Uplink**  
Streams encrypted telemetry data seamlessly over your factory's Wi-Fi using lightweight protocols.

- ESP32 EDGE
- MQTT PROTOCOL
- WI-FI / 4G



**STAGE 3 : ANALYZE**  
**AI Digital Twin**



**Stops Power Bleeding!**  
The AI engine instantly detects when machines are idling and wasting expensive electricity.



**Predicts Breakdowns**  
Uses ML to catch microscopic heat/vibration changes, warning you WEEKS before a motor burns out!

- AI RULE ENGINE
- TIMESERIES DB
- PRISMA ORM



**STAGE 4 : IMPACT**  
**Factory Outcomes**



**Incredible ROI!**  
See exact Rupee (₹) savings live on your dashboard as you eliminate idle power waste instantly!



**Smart Maintenance**  
Get instant App alerts saying: "Service Loom A today to prevent a breakdown tomorrow."

- NEXT.JS APP
- MOBILE ALERTS
- ZED GREEN